Vol-Arbitrage and Exotic Pricing on AETHER_CRYSTAL

IMC Prosperity Round 4 — Manual Trading Analysis

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Abstract

The simulator generates Aether Crystal under geometric Brownian motion with zero drift and annualised volatility $\sigma = 251\%$, yet every quoted option implies $\sigma_{\text{IV}} \approx 212\%$. The 39 vol-point gap is the source of all edge. We price the eleven instruments with closed-form Black–Scholes for vanillas and Monte Carlo for the three exotics (chooser, binary put, knock-out put), verify pricing identities (Rubinstein chooser decomposition; reflection-principle barrier formula), and solve a linear program for the maximum-EV portfolio under volume constraints. The recommended position — buy 50 contracts of every vanilla, the chooser, and the binary put — has expected PnL \$2.87M with standard deviation across the 100-simulation scoring of \$1.65M, giving a 97% probability of a positive score. Crucially, the seemingly-attractive knock-out put short (\$28,500 expected) is **rejected** after a left-tail correlation analysis: the KO position is positively correlated with the rest of the portfolio (Corr = +0.107), so its tail blow-ups occur on the same paths where the long-vol book is already underperforming. Including it improves the mean by \$27k but worsens the q_{01} score by \$48k and the absolute worst case by \$465k — a poor reward-to-risk trade-off.

1 Market Data and Parameters

The underlying obeys

$$dS_t = \sigma S_t \, dW_t, \qquad \sigma = 2.51, \quad S_0 = 50, \quad r = 0,$$

discretised at 4 steps per trading day with 252 trading days per year. We define $T_2 = 14/252 \approx 0.0556$ and $T_3 = 21/252 \approx 0.0833$. Note $\sigma\sqrt{T_3} = 0.7246$ — a regime where $N(d_1) \neq N(d_2)$ even at the money, so the standard “ATM call delta ≈ 0.5” intuition breaks down.

Contract	K	Type	T	Bid	Ask	BS Theo	Implied σ_{IV}
AC_50_P	50	put	T_3	12.00	12.05	14.143	2.121
AC_50_C	50	call	T_3	12.00	12.05	14.143	2.121
AC_35_P	35	put	T_3	4.33	4.35	5.870	2.122
AC_40_P	40	put	T_3	6.50	6.55	8.299	2.125
AC_45_P	45	put	T_3	9.05	9.10	11.072	2.119
AC_60_C	60	call	T_3	8.80	8.85	11.025	2.127
AC_50_P_2	50	put	T_2	9.70	9.75	11.631	2.089
AC_50_C_2	50	call	T_2	9.70	9.75	11.631	2.089

Table 1: Vanilla quotes, theoretical Black–Scholes values at $\sigma = 2.51$, and implied volatility from mid quotes.

Implied volatility is uniformly $\sigma_{\text{IV}} \approx 2.12$, against the true $\sigma = 2.51$. The gap is consistent across strikes and expiries, with the slightly lower IV at T_2 reflecting tiny variance loadings rather than true skew. *This is real edge, not a modelling error.*

2 Vanilla Pricing

Standard Black–Scholes with $r = 0$:

$$C(S, K, T, \sigma) = S\,N(d_1) - K\,N(d_2), \qquad P(S, K, T, \sigma) = K\,N(-d_2) - S\,N(-d_1),$$

where $d_1 = \frac{\ln(S/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$. Per-contract expected PnL on the buy side is $\mathbb{E}[\pi_{\text{buy}}] = 3000 \cdot (\mathbb{E}[\text{payoff}] - \text{ask})$, ranging from \$4,539 (AC_35_P) to \$6,533 (AC_60_C). All eight vanillas are buy-side +EV.

3 Exotic Pricing

3.1 Chooser Option AC_50_CO

The buyer chooses at T_2 whichever of call or put is in-the-money, then settles at T_3:

$$\Pi_{\mathrm{CO}} = \mathbf{1}_{S_{T_2}>K} \cdot \max(S_{T_3} - K, 0) + \mathbf{1}_{S_{T_2}\le K} \cdot \max(K - S_{T_3}, 0).$$

By Rubinstein's chooser identity (with $r = 0$):

$$\boxed{V_{\mathrm{CO}} = C(S_0, K, T_3) + P(S_0, K, T_2)}$$

Substituting: $V_{\mathrm{CO}} = 14.143 + 11.631 = 25.774$. Monte Carlo agrees to within 0.05. The bid is 22.20, ask is 22.30, giving edge $25.77 - 22.30 = \$3.47$ per unit, i.e. \$10,356 per contract — the largest single-trade edge in the round.

3.2 Binary Put AC_40_BP

Pays \$10 if $S_{T_3} < 40$:

$$V_{\mathrm{BP}} = 10 \cdot \Pr(S_{T_3} < 40) = 10 \cdot N(-d_2), \qquad d_2 = \frac{\ln(50/40) - \frac{1}{2}\sigma^2 T_3}{\sigma\sqrt{T_3}} = -0.0545,$$

giving $V_{\mathrm{BP}} = 10 \cdot N(0.0545) = 5.217$. Edge over ask \$5.10 is \$0.117 per unit (\$351 per contract). The position is *bounded* — max loss per contract is $5.10 \cdot 3000 = \$15{,}300$. Worth including.

3.3 Knock-Out Put AC_45_KO — Pricing

A down-and-out put with strike $K = 45$, barrier $B = 35$, expiry T_3. Under GBM with $r = 0$ the closed-form (Reiner–Rubinstein) is

$$V_{\mathrm{KO}} = P(S_0, K, T_3) - \left(\frac{B}{S_0}\right)^{2\lambda-2} P\left(\frac{B^2}{S_0}, K, T_3\right) - (K - B)\left[N(d_3) - \left(\frac{B}{S_0}\right)^{2\lambda-2} N(d_4)\right]$$

with $\lambda = 1 + (r - \frac{1}{2}\sigma^2)/\sigma^2 \approx 0.5$. Direct Monte Carlo gives $V_{\mathrm{KO}} \approx \$0.128$ vs. ask \$0.175 (buy $-$EV) and bid \$0.150 (sell +EV by \$0.022 per unit, i.e. \$57 per contract, \$28,500 at full size 500). The barrier is breached in $\Pr(\min_t S_t < 35) = 69.2\%$ of paths, so most of the time the option is dead-on-arrival.

4 Portfolio Optimisation

4.1 Linear-Programming Formulation

Variables $q_i \ge 0$ for each contract on its +EV side, with bounds $q_i \le V_i$ (volume cap). Objective:

$$\max_{\mathbf{q}} \sum_i q_i \cdot 3000 \cdot \mathrm{edge}_i.$$

With no risk constraint, the unconstrained LP returns $q_i = V_i$ for all +EV trades.

4.2 Why Static Delta-Hedging is Dominated

Adding a linear delta constraint $|\sum_i q_i \delta_i| \le 0$ trims $q_{\mathrm{AC\_50\_C\_2}}$ from 50 to 21 (and uses 200 short underlying as hedge), yielding $\mathbb{E}[\mathrm{PnL}] = \$2{,}696{,}278$ — a sacrifice of \$162k versus the unconstrained LP. But the gain in PnL standard deviation is only \$16.7M $\to$ \$15.5M, a 7% reduction. This makes static delta-hedging strictly dominated: variance is overwhelmingly driven by gamma/convexity, not first-order delta. The intuition: at $\sigma\sqrt{T_3} = 0.7246$, the price distribution at expiry is so dispersed that linear delta is a near-irrelevant component of the option value's path-dependence. **Recommendation: do not delta-hedge.**

4.3 The KO Short Trap: Positive Correlation Kills the Trade

The +\$28,500 EV from selling 500 KO puts is mathematically real, but its tail behaviour is correlated with the rest of the book in the *wrong* direction. We compute (over 500,000 simulated paths):

$$\mathrm{Corr}\big(\Pi_{\text{KO short}},\ \Pi_{\text{rest of book}}\big) = +0.107.$$

A negatively correlated KO position would offset losses elsewhere; a positive correlation *amplifies* them. Why does this happen?

Decomposition of the bad-KO paths. The KO short loses money when the path stays above \$35 throughout $[0, T_3]$ *and* ends below \$45 — i.e. when the underlying meanders without ever making a large excursion. These "low realised vol" paths are *also* the worst paths for the long-vol book, since every vanilla and the chooser is a long-vol position priced under $\sigma_{\text{IV}} \approx 2.12$ but valued at $\sigma = 2.51$. Without realised vol, the long-vol leg under-performs.

Conditional analysis. Conditioning on the 0.1% worst KO paths ($N = 500$):

Quantity	Conditional Mean	Unconditional Mean
KO-short PnL	−\$14,188,483	+\$26,934
Rest-of-book PnL	−\$ 6,567,666	+\$2,873,387
Total portfolio PnL	−\$20,756,149	+\$2,900,321

The rest of the book underperforms its mean by \$9.4M precisely on the paths where the KO short is blowing up. This is the opposite of a hedge.

4.4 Tournament-Score Impact of Including the KO Short

The score is the average PnL over $M = 100$ underlying paths. A naive "100-path averaging smooths everything" argument would suggest the KO tail risk is harmless. The actual tournament-score statistics across 10,000 simulated tournaments tell a different story:

Metric	With KO Short	Without KO Short	Δ
Mean score	\$2,900,528	\$2,873,774	+\$26,754
Std of score	\$1,671,437	\$1,652,516	+\$18,921
q_{01}	−\$ 608,179	−\$ 560,188	−\$47,991
q_{05}	+\$ 301,697	+\$ 308,617	−\$ 6,919
Worst case	−\$2,641,705	−\$2,176,915	−\$464,790
Pr(score < 0)	2.98%	2.90%	+0.08pp

Verdict: drop the KO short. The trade adds \$27k to the mean but degrades *every* downside metric: q_{01} worsens by \$48k, the absolute worst case worsens by \$465k, and Pr(score < 0) ticks up. The reward-to-tail-risk ratio is unattractive, especially for a one-shot tournament where minimising the chance of catastrophic underperformance matters more than optimising the last basis point of expected value.

Why 100-path averaging fails to smooth. A single-path KO blow-up is up to \$14.8M. Across 100 paths, that single event contributes \$148k to the score average. The probability of at least one tail-KO event in 100 paths is approximately $1 - (1 - 0.001)^{100} \approx 9.5\%$. So nearly 1-in-10 tournaments will see at least one blow-up that drags the score down. Combined with the positive correlation in Section 4.3, this is enough to overwhelm the small per-contract edge.

5 Final Recommendation

Contract	Action	Qty	Price	Contract EV
AC_50_P	BUY	50	12.05 ask	$6,233
AC_50_C	BUY	50	12.05 ask	$6,288
AC_35_P	BUY	50	4.35 ask	$4,539
AC_40_P	BUY	50	6.55 ask	$5,215
AC_45_P	BUY	50	9.10 ask	$5,874
AC_60_C	BUY	50	8.85 ask	$6,533
AC_50_P_2	BUY	50	9.75 ask	$5,662
AC_50_C_2	BUY	50	9.75 ask	$5,571
AC_50_CO	BUY	50	22.30 ask	$10,356
AC_40_BP	BUY	50	5.10 ask	$333
AC_45_KO	**NO TRADE**	0	—	$0
AC underlying	no trade	0	—	$0
			Total expected score	$2,873,387
			Score std (across 100-sim avg)	$1,652,516
			q_{05} **score**	$308,617
			Pr(**score** > 0)	97.1%

6 Lessons Applied from the Previous Round

The Round 3 auction post-mortem identified four weaknesses: prior miscalibration, no fixed-point Nash check, no asymmetric-incentive analysis, and no minimax-regret evaluation. The analogous failure modes in this round are mitigated as follows.

1. **Pricing-model verification.** Black–Scholes vs. Monte Carlo agree to < 0.1% on all vanillas. The chooser identity $V_{\text{CO}} = C(K, T_3) + P(K, T_2)$ is verified analytically and by MC.
2. **Stress-testing the actual scoring rule.** Simulating 10,000 independent 100-path tournaments produces the true score distribution rather than relying on per-path PnL. Without this, the 49% per-path loss probability would look alarming; with it, the 97% positive-score probability emerges.
3. **Correlation-aware tail-risk gating.** Initial reasoning treated the KO short as "small EV, isolated tail risk." The correlation analysis (Section 4.3) revealed Corr = +0.107 with the rest of the book. Tail risk is not isolated when the underlying-path conditioning is shared with the main position.
4. **Reward-vs-tail evaluation.** Explicitly compared score distributions with and without the KO short across 10,000 tournaments rather than relying on expected-value comparisons. The $27k mean gain is overwhelmed by the $48k–$465k worsening of every downside percentile.
5. **Hedging-instrument cost-benefit analysis.** Evaluated static delta-hedging via underlying short and confirmed it sacrifices $162k EV for only 7% standard deviation reduction — gamma is the dominant variance contributor at $\sigma\sqrt{T} \approx 0.7$.